

Exam Cheating Detection Report

27 Potential Cheating Incident(s) Detected

Video Duration	61m 24s
Original Resolution	1920x1080
Frame Rate	25.0 fps
Analysis Date	2026-02-04 13:02

Executive Summary

Analysis Overview:

- Total students identified: 22
- Teacher identified: Yes
- Potential cheating pairs detected: 27

Alert Summary

Students	Confidence	Tier	Primary Behaviors
R2_C5 & R2_C7	84.8%	HIGH	Body leaning toward peer (24700 instances), Han...
R2_C12 & R2_C10	78.7%	HIGH	Body leaning toward peer (18750 instances), Han...
R2_C6 & R3_C1	74.0%	HIGH	Body leaning toward peer (800 instances), Moved...
R3_C2 & R2_C6	73.5%	HIGH	Body leaning toward peer (550 instances), Hand ...
R2_C11 & R2_C10	69.0%	MEDIUM	Body leaning toward peer (12200 instances), Han...
R2_C12 & R2_C11	68.3%	MEDIUM	Body leaning toward peer (18500 instances), Han...
R2_C7 & R2_C8	67.6%	MEDIUM	Body leaning toward peer (3250 instances), Hand...
R2_C6 & R2_C8	67.1%	MEDIUM	Body leaning toward peer (800 instances), Hand ...
R2_C4 & R3_C1	64.6%	MEDIUM	Body leaning toward peer (650 instances), Moved...
R2_C4 & R2_C5	63.4%	MEDIUM	Body leaning toward peer (600 instances), Hand ...
R2_C10 & R2_C7	63.2%	MEDIUM	Body leaning toward peer (17000 instances), Han...
R2_C4 & R2_C6	63.0%	MEDIUM	Body leaning toward peer (100 instances), Hand ...
R2_C5 & R2_C8	62.0%	MEDIUM	Body leaning toward peer (2950 instances), Hand...
R2_C14 & R2_C13	61.8%	MEDIUM	Body leaning toward peer (2300 instances), Hand...
R2_C2 & R3_C1	61.2%	MEDIUM	Body leaning toward peer (5750 instances), Hand...
R3_C4 & R2_C14	61.2%	MEDIUM	Body leaning toward peer (2250 instances), Hand...
R2_C7 & R2_C6	59.9%	MEDIUM	Body leaning toward peer (800 instances), Hand ...
R2_C5 & R2_C6	59.8%	MEDIUM	Body leaning toward peer (750 instances), Hand ...
R2_C12 & R3_C3	59.3%	MEDIUM	Body leaning toward peer (1250 instances), Hand...
R2_C12 & R4_C2	57.3%	MEDIUM	Body leaning toward peer (150 instances), Hand ...

R2_C11 & R2_C14	57.0%	MEDIUM	Body leaning toward peer (6600 instances), Hand...
R2_C9 & R2_C8	56.1%	MEDIUM	Body leaning toward peer (200 instances), Hand ...
R2_C4 & R3_C2	54.8%	LOW	Body leaning toward peer (300 instances), Susta...
R2_C11 & R2_C13	52.2%	LOW	Body leaning toward peer (2050 instances), Hand...
R2_C10 & R2_C13	51.4%	LOW	Body leaning toward peer (2850 instances), Hand...
R2_C11 & R3_C3	50.6%	LOW	Body leaning toward peer (800 instances), Hand ...
R2_C11 & R2_C8	50.0%	LOW	Body leaning toward peer (3600 instances), Hand...

Detailed Alert Analysis

Alert 1: R2_C5 & R2_C7

Confidence Score: 84.8% (HIGH)

Suspicious Time Periods: 518.2s - 528.0s, 742.4s - 752.2s, 516.2s - 526.0s, 520.2s - 532.0s, 744.4s - 754.2s

Detected Behaviors:

- Body leaning toward peer (24700 instances)
 - Hand activity near face (24695 instances)
 - Mutual gaze (looking at each other) (9255 instances)
 - Sustained head turn toward peer (6747 instances)
 - Whispering posture detected (6727 instances)
 - Moved closer than normal (210 instances)
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Alert 2: R2_C12 & R2_C10

Confidence Score: 78.7% (HIGH)

Suspicious Time Periods: 992.4s - 1006.0s, 994.8s - 1008.0s, 990.4s - 1003.8s, 1002.0s - 1012.0s, 985.8s - 999.0s

Detected Behaviors:

- Body leaning toward peer (18750 instances)
 - Hand activity near face (18750 instances)
 - Sustained head turn toward peer (14721 instances)
 - Whispering posture detected (14636 instances)
 - Moved closer than normal (7930 instances)
 - Mutual gaze (looking at each other) (4264 instances)
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Alert 3: R2_C6 & R3_C1

Confidence Score: 74.0% (HIGH)

Suspicious Time Periods: 94.0s - 104.2s, 96.4s - 106.2s, 91.6s - 102.2s, 89.6s - 100.2s, 98.4s - 118.4s

Detected Behaviors:

- Body leaning toward peer (800 instances)
- Moved closer than normal (800 instances)
- Hand activity near face (683 instances)
- Mutual gaze (looking at each other) (474 instances)
- Sustained head turn toward peer (301 instances)
- Whispering posture detected (194 instances)

Alert 4: R3_C2 & R2_C6

Confidence Score: 73.5% (HIGH)

Suspicious Time Periods: 96.8s - 119.4s, 94.8s - 106.6s, 92.2s - 104.0s, 90.2s - 101.0s, 11.4s - 85.6s

Detected Behaviors:

- Body leaning toward peer (550 instances)
 - Hand activity near face (408 instances)
 - Sustained head turn toward peer (384 instances)
 - Whispering posture detected (273 instances)
 - Mutual gaze (looking at each other) (226 instances)
 - Moved closer than normal (144 instances)
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Alert 5: R2_C11 & R2_C10

Confidence Score: 69.0% (MEDIUM)

Suspicious Time Periods: 35.4s - 45.2s, 716.6s - 732.0s, 714.4s - 729.4s, 936.0s - 950.2s, 718.6s - 734.4s

Detected Behaviors:

- Body leaning toward peer (12200 instances)
 - Hand activity near face (12200 instances)
 - Sustained head turn toward peer (10150 instances)
 - Whispering posture detected (10100 instances)
 - Moved closer than normal (7625 instances)
 - Mutual gaze (looking at each other) (1836 instances)
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Alert 6: R2_C12 & R2_C11

Confidence Score: 68.3% (MEDIUM)

Suspicious Time Periods: 454.6s - 470.8s, 172.2s - 184.0s, 170.0s - 181.8s, 174.2s - 187.6s, 178.0s - 189.6s

Detected Behaviors:

- Body leaning toward peer (18500 instances)
- Hand activity near face (18490 instances)
- Sustained head turn toward peer (16870 instances)
- Whispering posture detected (16435 instances)
- Moved closer than normal (5308 instances)
- Mutual gaze (looking at each other) (1740 instances)

Alert 7: R2_C7 & R2_C8

Confidence Score: 67.6% (MEDIUM)

Suspicious Time Periods: 90.2s - 106.6s, 97.0s - 108.8s, 88.0s - 103.4s, 99.0s - 110.8s, 86.0s - 101.4s

Detected Behaviors:

- Body leaning toward peer (3250 instances)
 - Hand activity near face (3250 instances)
 - Sustained head turn toward peer (1064 instances)
 - Whispering posture detected (1064 instances)
 - Mutual gaze (looking at each other) (940 instances)
 - Moved closer than normal (1 instances)
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Alert 8: R2_C6 & R2_C8

Confidence Score: 67.1% (MEDIUM)

Suspicious Time Periods: 81.4s - 94.0s, 85.6s - 96.8s, 87.6s - 98.8s, 46.4s - 91.6s, 44.4s - 89.4s

Detected Behaviors:

- Body leaning toward peer (800 instances)
 - Hand activity near face (800 instances)
 - Sustained head turn toward peer (557 instances)
 - Whispering posture detected (414 instances)
 - Mutual gaze (looking at each other) (374 instances)
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Alert 9: R2_C4 & R3_C1

Confidence Score: 64.6% (MEDIUM)

Suspicious Time Periods: 36.0s - 60.8s, 33.4s - 54.0s, 26.6s - 47.6s, 21.8s - 40.4s, 38.4s - 63.4s

Detected Behaviors:

- Body leaning toward peer (650 instances)
 - Moved closer than normal (627 instances)
 - Sustained head turn toward peer (579 instances)
 - Hand activity near face (546 instances)
 - Whispering posture detected (281 instances)
 - Mutual gaze (looking at each other) (179 instances)
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Alert 10: R2_C4 & R2_C5

Confidence Score: 63.4% (MEDIUM)

Suspicious Time Periods: 3.8s - 40.4s, 32.2s - 44.8s, 25.0s - 42.6s, 35.4s - 46.8s, 40.6s - 51.4s

Detected Behaviors:

- Body leaning toward peer (600 instances)
 - Hand activity near face (600 instances)
 - Sustained head turn toward peer (503 instances)
 - Whispering posture detected (464 instances)
 - Moved closer than normal (389 instances)
 - Mutual gaze (looking at each other) (80 instances)
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Alert 11: R2_C10 & R2_C7

Confidence Score: 63.2% (MEDIUM)

Suspicious Time Periods: 894.4s - 904.2s, 707.0s - 717.4s, 709.0s - 719.4s, 1192.2s - 1206.2s, 896.4s - 906.2s

Detected Behaviors:

- Body leaning toward peer (17000 instances)
 - Hand activity near face (17000 instances)
 - Mutual gaze (looking at each other) (9138 instances)
 - Moved closer than normal (3046 instances)
 - Sustained head turn toward peer (2226 instances)
 - Whispering posture detected (2216 instances)
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Alert 12: R2_C4 & R2_C6

Confidence Score: 63.0% (MEDIUM)

Suspicious Time Periods: 11.4s - 43.8s, 19.4s - 46.0s

Detected Behaviors:

- Body leaning toward peer (100 instances)
 - Hand activity near face (97 instances)
 - Moved closer than normal (85 instances)
 - Sustained head turn toward peer (84 instances)
 - Whispering posture detected (57 instances)
 - Mutual gaze (looking at each other) (26 instances)
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Alert 13: R2_C5 & R2_C8

Confidence Score: 62.0% (MEDIUM)

Suspicious Time Periods: 95.8s - 107.8s, 93.4s - 104.6s, 98.2s - 110.0s, 118.6s - 173.2s, 91.4s - 102.6s

Detected Behaviors:

- Body leaning toward peer (2950 instances)
 - Hand activity near face (2950 instances)
 - Sustained head turn toward peer (937 instances)
 - Whispering posture detected (937 instances)
 - Mutual gaze (looking at each other) (581 instances)
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Alert 14: R2_C14 & R2_C13

Confidence Score: 61.8% (MEDIUM)

Suspicious Time Periods: 3.8s - 34.0s, 74.6s - 88.4s, 76.6s - 90.8s, 79.4s - 92.8s, 10.4s - 36.0s

Detected Behaviors:

- Body leaning toward peer (2300 instances)
 - Hand activity near face (2300 instances)
 - Sustained head turn toward peer (2064 instances)
 - Whispering posture detected (2064 instances)
 - Mutual gaze (looking at each other) (130 instances)
 - Moved closer than normal (86 instances)
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Alert 15: R2_C2 & R3_C1

Confidence Score: 61.2% (MEDIUM)

Suspicious Time Periods: 194.4s - 205.0s, 463.8s - 474.8s, 200.4s - 218.4s, 192.2s - 202.4s, 196.4s - 207.0s

Detected Behaviors:

- Body leaning toward peer (5750 instances)
 - Hand activity near face (5740 instances)
 - Sustained head turn toward peer (4597 instances)
 - Whispering posture detected (2906 instances)
 - Moved closer than normal (1332 instances)
 - Mutual gaze (looking at each other) (1252 instances)
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Alert 16: R3_C4 & R2_C14

Confidence Score: 61.2% (MEDIUM)

Suspicious Time Periods: 4.6s - 19.6s, 21.8s - 33.6s, 23.8s - 35.6s, 25.8s - 37.6s, 27.8s - 39.6s

Detected Behaviors:

- Body leaning toward peer (2250 instances)
 - Hand activity near face (2170 instances)
 - Sustained head turn toward peer (2105 instances)
 - Whispering posture detected (1908 instances)
 - Mutual gaze (looking at each other) (82 instances)
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Alert 17: R2_C7 & R2_C6

Confidence Score: 59.9% (MEDIUM)

Suspicious Time Periods: 22.8s - 45.4s, 19.4s - 42.4s, 11.4s - 38.0s, 30.4s - 52.2s, 32.8s - 82.0s

Detected Behaviors:

- Body leaning toward peer (800 instances)
 - Hand activity near face (794 instances)
 - Sustained head turn toward peer (417 instances)
 - Mutual gaze (looking at each other) (392 instances)
 - Whispering posture detected (270 instances)
 - Moved closer than normal (144 instances)
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Alert 18: R2_C5 & R2_C6

Confidence Score: 59.8% (MEDIUM)

Suspicious Time Periods: 19.6s - 45.8s, 32.2s - 82.2s, 23.6s - 67.8s, 39.4s - 84.2s, 68.0s - 90.2s

Detected Behaviors:

- Body leaning toward peer (750 instances)
 - Hand activity near face (750 instances)
 - Mutual gaze (looking at each other) (409 instances)
 - Sustained head turn toward peer (331 instances)
 - Whispering posture detected (213 instances)
 - Moved closer than normal (182 instances)
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Alert 19: R2_C12 & R3_C3

Confidence Score: 59.3% (MEDIUM)

Suspicious Time Periods: 40.2s - 61.6s, 36.6s - 59.6s, 31.6s - 50.0s, 45.4s - 64.2s, 64.4s - 79.4s

Detected Behaviors:

- Body leaning toward peer (1250 instances)
- Hand activity near face (1198 instances)

- Sustained head turn toward peer (830 instances)
 - Whispering posture detected (717 instances)
 - Moved closer than normal (409 instances)
 - Mutual gaze (looking at each other) (82 instances)
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Alert 20: R2_C12 & R4_C2

Confidence Score: 57.3% (MEDIUM)

Suspicious Time Periods: 11.8s - 33.2s, 5.8s - 26.4s, 3.8s - 22.6s

Detected Behaviors:

- Body leaning toward peer (150 instances)
 - Hand activity near face (141 instances)
 - Sustained head turn toward peer (135 instances)
 - Whispering posture detected (101 instances)
 - Moved closer than normal (63 instances)
 - Mutual gaze (looking at each other) (11 instances)
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Alert 21: R2_C11 & R2_C14

Confidence Score: 57.0% (MEDIUM)

Suspicious Time Periods: 64.6s - 75.8s, 61.0s - 73.6s, 47.0s - 56.8s, 66.6s - 78.6s, 45.0s - 54.8s

Detected Behaviors:

- Body leaning toward peer (6600 instances)
 - Hand activity near face (6600 instances)
 - Mutual gaze (looking at each other) (1183 instances)
 - Sustained head turn toward peer (575 instances)
 - Whispering posture detected (575 instances)
 - Moved closer than normal (280 instances)
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Alert 22: R2_C9 & R2_C8

Confidence Score: 56.1% (MEDIUM)

Suspicious Time Periods: 17.2s - 38.8s, 10.0s - 24.2s, 20.2s - 42.4s, 15.2s - 35.8s

Detected Behaviors:

- Body leaning toward peer (200 instances)
- Hand activity near face (195 instances)
- Sustained head turn toward peer (149 instances)
- Whispering posture detected (106 instances)
- Mutual gaze (looking at each other) (63 instances)

- Moved closer than normal (3 instances)
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Alert 23: R2_C4 & R3_C2

Confidence Score: 54.8% (LOW)

Suspicious Time Periods: 45.0s - 62.4s, 11.4s - 52.4s, 30.2s - 60.0s, 47.8s - 65.6s, 17.2s - 55.6s

Detected Behaviors:

- Body leaning toward peer (300 instances)
 - Sustained head turn toward peer (291 instances)
 - Hand activity near face (254 instances)
 - Moved closer than normal (190 instances)
 - Whispering posture detected (88 instances)
 - Mutual gaze (looking at each other) (39 instances)
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Alert 24: R2_C11 & R2_C13

Confidence Score: 52.2% (LOW)

Suspicious Time Periods: 68.0s - 77.8s, 47.8s - 60.8s, 70.0s - 79.8s, 72.0s - 81.8s, 52.2s - 65.4s

Detected Behaviors:

- Body leaning toward peer (2050 instances)
 - Hand activity near face (2050 instances)
 - Mutual gaze (looking at each other) (713 instances)
 - Sustained head turn toward peer (391 instances)
 - Whispering posture detected (391 instances)
 - Moved closer than normal (3 instances)
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Alert 25: R2_C10 & R2_C13

Confidence Score: 51.4% (LOW)

Suspicious Time Periods: 155.0s - 165.0s, 151.0s - 161.0s, 45.8s - 58.2s, 48.2s - 61.2s, 50.2s - 63.2s

Detected Behaviors:

- Body leaning toward peer (2850 instances)
- Hand activity near face (2850 instances)
- Mutual gaze (looking at each other) (1447 instances)
- Moved closer than normal (745 instances)
- Sustained head turn toward peer (458 instances)
- Whispering posture detected (458 instances)

Alert 26: R2_C11 & R3_C3

Confidence Score: 50.6% (LOW)

Suspicious Time Periods: 72.6s - 86.4s, 69.4s - 84.4s, 42.6s - 62.4s, 37.0s - 59.8s, 67.2s - 82.4s

Detected Behaviors:

- Body leaning toward peer (800 instances)
 - Hand activity near face (785 instances)
 - Mutual gaze (looking at each other) (597 instances)
 - Sustained head turn toward peer (114 instances)
 - Whispering posture detected (41 instances)
 - Moved closer than normal (16 instances)
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Alert 27: R2_C11 & R2_C8

Confidence Score: 50.0% (LOW)

Suspicious Time Periods: 131.0s - 142.8s, 133.0s - 145.0s, 135.0s - 147.0s, 137.2s - 149.0s, 141.0s - 151.2s

Detected Behaviors:

- Body leaning toward peer (3600 instances)
- Hand activity near face (3600 instances)
- Mutual gaze (looking at each other) (2187 instances)
- Sustained head turn toward peer (8 instances)
- Whispering posture detected (8 instances)

Methodology

This analysis was conducted using a 7-stage computer vision pipeline:

1. Video Preprocessing

Frames extracted at 5 fps with CLAHE contrast enhancement in LAB color space.

2. Person Detection & Tracking

YOLOv8 object detection with BoT-SORT multi-object tracking for consistent student identification throughout the video.

3. Pose Estimation

YOLOv8-Pose model extracts 17 COCO body keypoints. Head orientation is computed using solvePnP with facial landmarks.

4. Spatial Analysis

Automatic seat assignment based on median positions during warm-up period. Neighbor graphs built using distance thresholds. Teachers identified by movement patterns and standing posture.

5. Behavioral Feature Extraction

Per-frame features include: head direction, mutual gaze detection, whispering posture, body lean, hand position relative to face and neighbors, paper sharing gestures.

6. Temporal Classification

Sliding window analysis (10-second windows, 2-second stride) with weighted scoring. Multi-criteria filter requires: minimum suspicious windows, confidence threshold, mutual behavior evidence, and absence of nearby teacher.

7. Alert Generation

Confidence tiers (Very High > 85%, High > 70%, Medium > 55%) with timestamped evidence and behavior summaries.

Important Notes:

- This system is designed to flag potential incidents for human review.
- False positives may occur due to normal classroom interactions.
- Results should be verified by reviewing the evidence frames and video.
- Teacher proximity during flagged periods may indicate supervised activity.